

$$X(z) = \sum_{n=0}^{\infty} n 2^n z^{-n} = \sum_{n=0}^{\infty} n \left(\frac{2}{z}\right)^n$$

~~scribbles~~  $\frac{2}{z} = r$

$$\rightarrow X^+(z) = \sum_{n=0}^{\infty} n r^n$$

$$S = 0 + r + 2r^2 + 3r^3 + \dots$$

$$rS = 0 + r^2 + 2r^3 + 3r^4 + \dots$$

$$S(1-r) = r + r^2 + r^3 + \dots$$

$$S(1-r) = \frac{r}{1-r}$$

$$S = \frac{r}{(1-r)^2}$$

$$\rightarrow X^+(z) = \frac{\frac{2}{z}}{\left(1 - \frac{2}{z}\right)^2} \Rightarrow \frac{2z^2}{z(z-2)^2}$$

$$X^+(z) = \frac{2z}{(z-2)^2}$$